

RAHID KHAN

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

Artificial Intelligence undergraduate (BS, expected 2027) with hands-on experience building and deploying machine learning systems in Python, PyTorch, and TensorFlow. Practical exposure to computer vision (CNN/Vision Transformer models), NLP, and LLM-based agent systems, including a research automation agent built on the Model Context Protocol (MCP) integrating LangChain and Groq LLM inference. Comfortable across the full pipeline — data preprocessing, model training and evaluation, and deployment via Flask/FastAPI web applications. Seeking an AI/ML Engineering or GenAI Engineering role to apply and grow these skills in production settings.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C++, SQL (Basic)

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNN Architectures, Vision Transformers (ViT), Grad-CAM, Model Fine-Tuning, Transfer Learning

LLMs & Generative AI: LangChain, LLM Agents, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Prompt Engineering, Groq LLM APIs (Llama 3.3/4)

Model Deployment & APIs: Flask, FastAPI, REST APIs, Docker (containerized deployment)

Data Processing & Visualization: Pandas, NumPy, Matplotlib, Data Preprocessing, Feature Engineering

Databases & Vector Stores: SQLAlchemy, SQLite, FAISS

Tools & Platforms: Git, GitHub, Tkinter, Linux, Jupyter Notebook

PROJECTS

MCP-Based AI Research Agent | *LangChain · Groq LLM · Flask · Model Context Protocol*

- Built a full-stack research automation agent that connects an LLM (Groq-hosted Llama 3.3 70B / Llama 4) to 44+ external tools across five MCP servers covering ArXiv, Semantic Scholar, PubMed, CrossRef, and web search.
- Designed a three-panel Flask web interface with real-time streaming of the agent's reasoning and tool-execution logs, session history, and configurable temperature/model settings.
- Implemented failover logic to auto-switch search tools when rate-limited, plus automatic PDF retrieval and DOI/ArXiv citation extraction; packaged for both local (Flask) and production (Gunicorn, Docker) deployment.

Eye Disease Classification System | *PyTorch · TensorFlow · Vision Transformer · Flask*

- Developed an image classification pipeline to detect eye diseases from retinal images, covering dataset preparation, augmentation, and preprocessing.
- Trained and compared CNN and Vision Transformer (ViT) architectures, evaluating results with confusion matrices, ROC curves, and per-class classification reports.
- Applied Grad-CAM visualizations to interpret model predictions and deployed the trained model behind a lightweight Flask web application for interactive inference.

Multi-User RAG Web Platform | *FastAPI · LangChain · FAISS · SQLAlchemy*

- Built a full-stack, multi-user Retrieval-Augmented Generation platform where each user creates isolated knowledge bases, uploads documents, and chats with their own data.
- Implemented a multi-format ingestion pipeline (PDF, DOCX, TXT, MD, CSV, PPTX, code files, and URLs) with automatic chunking, embedding, and per-knowledge-base FAISS vector store indexing for zero data leakage between users.
- Designed a bring-your-own-key architecture supporting OpenAI, Anthropic, Groq, and Hugging Face LLMs, with provider keys protected using Fernet encryption at rest.
- Added JWT/Google OAuth authentication with role-based access control, real-time token-by-token chat streaming via Server-Sent Events with source citations, and a usage analytics dashboard.

EDUCATION

BS in Artificial Intelligence — CITY University of Science & Information Technology, Peshawar

Expected Graduation: 2027

CERTIFICATIONS

- AI & Machine Learning Training Program — NAVTTC (2024)

LANGUAGES

English, Urdu, Pashto